# LAB 3: UML Class Diagram Design

## Smart Waste Collection Management System

KIET Deemed To Be University | Computer Science Engineering | CSE-E

**Prepared by: Tanishka Singh - Leader (Product Manager)**

## 1. Objective

To identify the classes, attributes, methods and relationships from the requirements in the Software Requirements Specification (SRS v1.0) of the Smart Waste Collection Management System, and to represent them as a UML class diagram (Figure 1, last page).

## 2. Identified classes

| Class | Description | SRS reference |
|---|---|---|
| User (abstract) | Base class for every person who logs in; stores identity and credentials. | 2.3, REQ-DB-3, 5.3 |
| Administrator | Municipal administrator. Configures alert thresholds, requests and edits routes, views analytics. | 2.3, REQ-AL-1, REQ-RO-3, REQ-DB-2 |
| Driver | Collection staff using the mobile app. Views the assigned route and marks bins as collected. | 2.3, 3.1 |
| Citizen | Member of the public who reports overflowing bins and tracks the complaint status. | REQ-CF-1, REQ-CF-3 |
| Bin | Physical waste bin with location, fill-level, alert threshold and status. | 4.1, REQ-AL-1, 5.5 |
| SensorNode | Ultrasonic sensor with microcontroller that measures and transmits the fill-level. | 3.2, REQ-BM-1 to 3 |
| SensorReading | One timestamped fill-level measurement; builds the history used for analytics. | REQ-BM-2, REQ-DB-2 |
| Alert | Threshold alert sent to admin and driver; logged for audit. | REQ-AL-2, REQ-AL-3 |
| Route | Optimized collection route covering bins at or above the threshold. | REQ-RO-1 to 3 |
| Complaint | Citizen complaint with a unique tracking ID and Pending/Resolved status. | REQ-CF-1 to 3 |
| Report | Analytics report: number of collections and average fill-level over a date range. | REQ-DB-2 |

## 3. Relationships

| From | Type | To | Mult. | Meaning |
|---|---|---|---|---|
| Administrator, Driver, Citizen | Generalization | User | - | Each is a kind of User and inherits login()/logout(). |
| Bin | Association | SensorNode | 1 : 1 | Each bin is monitored by one sensor node. |
| SensorNode | Association | SensorReading | 1 : * | A node records many readings over time. |
| Bin | Association | Alert | 1 : * | A bin triggers an alert when fill-level crosses the threshold. |
| Alert | Dependency | User | * : 1..* | An alert notifies the admin and the relevant driver. |
| Route | Aggregation | Bin | 1 : * | A route includes many bins; bins exist without the route. |
| Administrator | Association | Route | 1 : * | Administrator generates and edits routes. |

| From | Type | To | Mult. | Meaning |
|---|---|---|---|---|
| Driver | Association | Route | 1 : * | Driver is assigned to routes. |
| Administrator | Association | Report | 1 : * | Administrator views analytics reports. |
| Citizen | Association | Complaint | 1 : * | A citizen submits many complaints. |
| Complaint | Association | Bin | * : 1 | Each complaint is reported on one bin. |

## 4. Design notes

- Bin.markCollected() also sets the linked Complaint to Resolved (business rule in SRS 5.5).
- Bin.isAboveThreshold() decides whether a bin is included in the next Route (REQ-RO-1, SRS 5.5).
- SensorNode.retryRead() supports error handling for failed sensor readings (REQ-BM-3).
- Alert.log() stores every alert with bin ID and timestamp for audit (REQ-AL-3).
- The System Administrator role from SRS 2.3 is not modelled as a class because it only maintains servers and hardware.

# UML Class Diagram - Smart Waste Collection Management System

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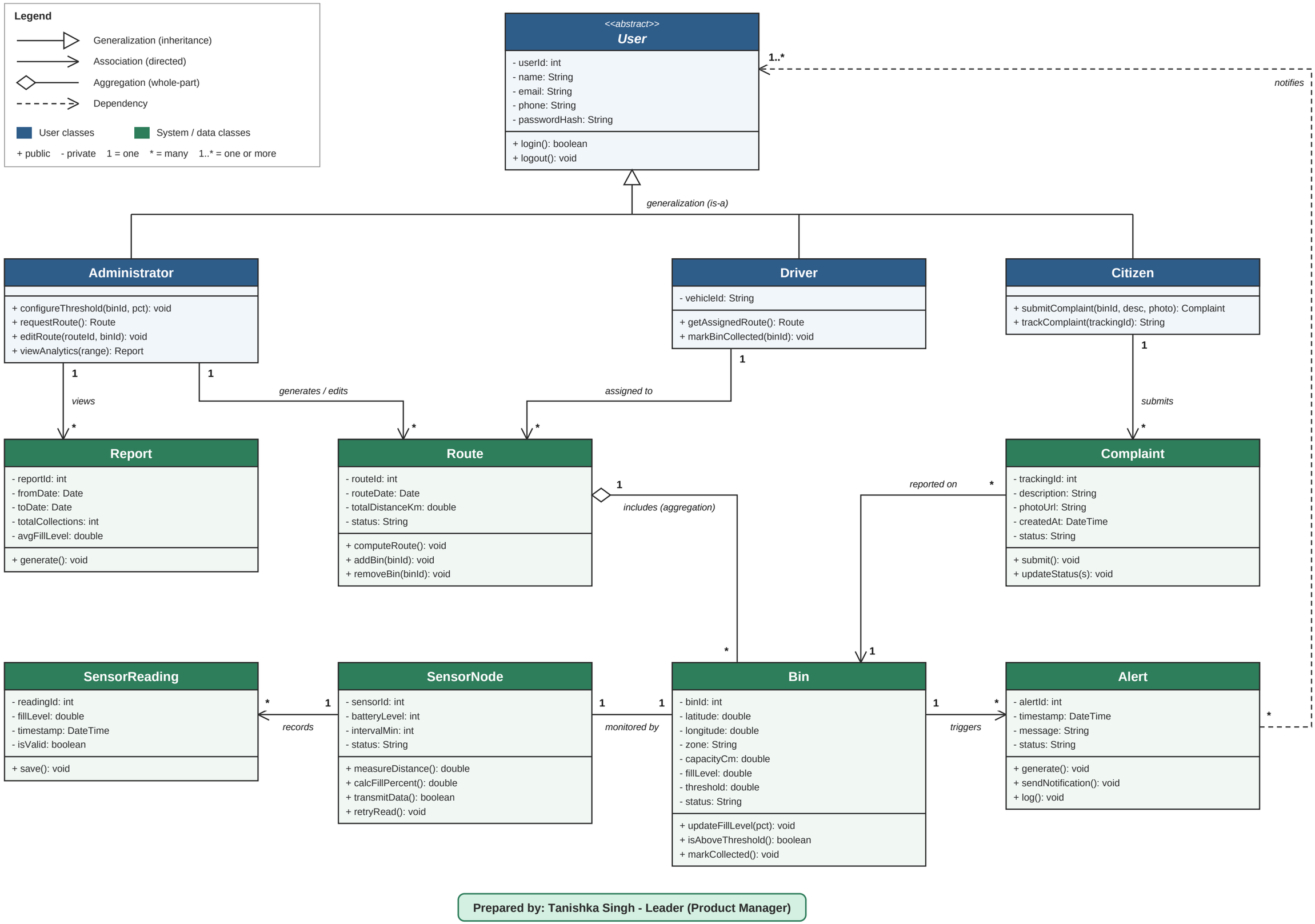


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